Dumindu Kumarapeli

+94 76 085 8628 | duminduku.25@gmail.com | Colombo, Sri Lanka
linkedin.com/in/dumindu-kumarapeli-7a4636309 | github.com/Dumee-25 | portfolio.kadpkumarapeli.workers.dev

Data Science undergraduate building LLM systems end to end (RAG, agents, local-model inference), with first-author research on privacy-preserving LLM analysis, epidemic forecasting and clinical prediction.

Education

NSBM Green University — Sri Lanka
BSc (Hons) Data Science, currently Year 3 — *2025 – Expected Dec 2028*

- GPA 3.85/4.00.

Research and Publications

- **D. Kumarapeli** and H. Weerasinghe, "Privacy-preserving automated exploratory data analysis using deterministic summarization and local large language models," *Proc. NSBM Student Research Symposium on Innovation Technology (NSRSIT)*, 2026, pp. 146–150. **Best Overall Research**, session award.
- **D. Kumarapeli** and H. Weerasinghe, "Timescale separation and model reduction in operational dengue forecasting," ICCMM 2026, University of Colombo. [accepted, oral presentation]
- **D. Kumarapeli** and H. Weerasinghe, "A regime audit for detecting constraint-artefact reproduction numbers in hybrid mechanistic–machine learning epidemic models." [under review]
- Extended NSRSIT paper on DataSense v2: measurable privacy, checked LLM output. Target: ICARC. [in preparation]
- **D. Kumarapeli** and H. Weerasinghe, "A hybrid compartmental and neural network model for district-level dengue forecasting in Sri Lanka: forecast skill, spatial heterogeneity and identifiability." Target: *Ceylon J. Sci.* [in preparation]
- **D. Kumarapeli** and H. Weerasinghe, "Proxy labels versus observed endpoints: an audit and benchmark of clinical benefit prediction on MSK-CHORD." [in preparation]

LLM and AI Projects

DataSense v2: Privacy-Preserving Automated EDA | *Polars, Presidio, OpenDP* — portfolio case study

- Rebuilt the NSRSIT system local-first: privacy tiers limit each prompt, Presidio finds personal data, optional differential privacy adds noise, and every number the model writes is checked against the facts it cites.
- Built a benchmark on the frozen engine: it found 4,644 of 5,421 problems planted in 50 OpenML tables, and 27 of the 35 columns it flagged in Raha's real dirty tables had wrong cells. It still misses typos.

OWL: Datacenter AI Operations Platform | *Gemini, MCP* — github.com/Dumee-25/OWL-Datacenter_agent

- Built an agentic RAG assistant over ChromaDB with hybrid vector and keyword retrieval and recency re-ranking, served through REST, WebSocket and a nine-tool MCP server.
- Paired an Isolation Forest and LSTM autoencoder ensemble with Gemini-generated root-cause reports; every alert carries a confidence score and the system never acts on infrastructure.
- Wrote a simulator producing 4.7M telemetry rows over 18 months; logged per-call LLM latency and cost.

Fulyn: Private Tool-Calling Agent | *pgvector, Ollama* — github.com/Dumee-25/Fulyn

- Agent turns free-text diary entries into linked expense, sleep, mood and journal records; every tool call is schema-checked and invalid arguments are returned to the model to retry.
- Replaced vector-only search after finding unrelated queries scored 0.62 similarity, higher than some true matches; now merges vector and full-text rankings and flags keyword-free hits.
- Ran opt-in tool-calling tests across local models; llama3.2 failed most by picking the wrong tool or inventing arguments.

Technical Skills

LLM / RAG: Retrieval-augmented generation (RAG), AI agents and tool calling, prompt engineering, Ollama, Gemini, OpenAI API, ChromaDB, pgvector, Model Context Protocol (MCP)
ML / Data: PyTorch, scikit-learn, XGBoost, LightGBM, pandas, Polars, NumPy, SciPy, YOLOv8, OpenCV
Privacy: Differential privacy (OpenDP), personal-data detection (Microsoft Presidio), prompt-injection filtering
Languages: Python, SQL, Java, JavaScript, TypeScript, R
Backend / Tools: FastAPI, PostgreSQL, SQLite, Docker, Git/GitHub, Linux, Next.js, React, CUDA
Spoken: English (fluent), Sinhala (native)

Awards

Best Overall Research, session award, NSRSIT 2026 (2026) | Data Storm 7.0 Final Round, Top 10, team Data Drifters (2026)